

Manasa Sambaraju

Senior QA Engineer | Data & GenAI Quality | AI Validation & RAG Systems |

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SUMMARY

Senior QA Engineer with **7.6 years of industry experience** in functional, data, and backend testing across **healthcare, banking, and IAM domains**, currently transitioning into **AI / Gen AI Quality & Validation**. Hands-on experience building and validating **Retrieval-Augmented Generation (RAG)** systems with a strong focus on **AI reliability, explainability, and evidence-grounded outputs**. Brings a **quality-first mindset** to AI systems, combining traditional QA rigor with emerging **LLM, data, and model validation practices** in regulated environments.

KEY SKILLS

Quality Engineering & Testing - Quality Assurance & Software Testing, Test Strategy & Test Planning , Functional, Integration & Regression Testing, Defect Management & Root Cause Analysis, Requirements Analysis & Test Design

AI / Gen AI Quality & Validation - AI / Gen AI Quality & Validation, Model Output Validation (Project-based), AI Reliability & Risk-Aware Testin Explainability & Evidence-Grounded Validation, Hallucination-Safe Response Validation

Data & Analytics Quality - Data Quality & Analytics Validation, SQL-Based Data Verification, Backend Data Consistency & Accuracy Checks

API & Backend Validation-Backend & API Validation (Manual),REST API Request / Response Analysis

Agile & Collaboration- Agile / Scrum QA Practices, Cross-Functional Collaboration, Stakeholder Communication

Documentation & Governance- Documentation & Test Reporting, Traceability & Audit-Ready Artifacts, Continuous Quality Improvement

TECHNICAL SKILLS

Programming & Querying - Python, SQL, Java

Generative AI & LLM Systems - Generative AI, Large Language Models (LLMs),Retrieval-Augmented Generation (RAG),Prompt Behavior Analysis & Regression, Model Output & Response Validation, Hallucination Detection & Evidence Grounding

Machine Learning - Supervised Learning, Model Evaluation Metrics, Feature Engineering & Data Preprocessing, Bias & Outcome Distribution Analysis

Data & Analytics-Data Validation(Accuracy, Completeness, Consistency),Backend Data Verification, Pandas, Exploratory Data Analysis

API & Backend Testing- REST APIs (Manual Testing),Postman, JSON Request / Response Validation

Vector Databases & Retrieval: FAISS, Embedding-Based Semantic Search ,Open AI Embeddings (Batched)

QA, Dev & Collaboration Tools :JIRA, Confluence, GitHub, Jupyter Notebook, Google Colab, VSCode

Cloud & Platforms (Foundational): Microsoft Azure Fundamentals (AZ-900), Azure Data Fundamentals (DP-900)

PROJECTS

Healthcare GenAI Claims Assistant | GenAI Decision-Support System | RAG | AI Validation Focus

- Built a **GenAI-powered healthcare claims decision-support system** using Retrieval-Augmented Generation (RAG)
- Implemented **semantic retrieval and evidence-grounded LLM responses** to ensure explainable, hallucination-safe decisions
- Designed **structured outputs** (coverage decision, conditions, evidence, confidence) for regulated healthcare use cases
- Focused on **AI reliability, explainability, and validation** using synthetic claims and public policy documents

Insurance Fraud Detection | Machine Learning, Python

- Developed **classification models** to detect fraudulent insurance claims
- Performed **feature analysis and model evaluation** to reduce false positives
- Applied **data validation and result analysis** to improve decision reliability

Sentiment-Based Product Recommendation | NLP, Python

- Built an NLP-driven recommendation system combining **sentiment analysis and collaborative filtering**
- Evaluated model outputs to improve recommendation relevance and accuracy

PROFESSIONAL EXPERIENCE

QA Analyst	Aug '22 - Jan '23
Tata Consultancy Services (TCS)	Bengaluru,Karnataka
• Supported QA activities for a large-scale banking application requiring high data accuracy and process reliability • Performed SQL-based data validation and backend checks to ensure data integrity across workflows • Collaborated with cross-functional Agile teams to improve testing efficiency and reporting visibility • Contributed to process improvements that reduced turnaround time and improved defect resolution metrics	
Test Analyst	Jan '18 - Jan '22
Infosys	Hyderabad,Telangana

- Led end-to-end testing for **IAM and healthcare applications**, ensuring regulatory compliance and system stability
- Designed and executed **functional, regression, integration, and manual API tests**
- Performed backend data validation using SQL to verify system workflows and business rules
- Mentored junior testers and contributed to QA best practices across Agile teams
- Reduced production defect leakage and improved overall release quality through structured QA processes

Test Engineer

Feb '15 - Dec '17

Infosys

Hyderabd,Telangana

- Tested healthcare applications related to **member enrollment and claims processing**
- Collaborated with BAs and developers to analyze requirements and design comprehensive test cases
- Executed functional, integration, and regression testing using **ALM** and **JIRA**
- Identified critical defects through detailed analysis, improving release stability

EDUCATION

Post Graduate Diploma in Data Science & AI (NLP Specialization)

Sep '24 - Oct '25

International Institute of Information Technology (IIIT-Bangalore)

Bengaluru,Karnataka

Bachelor of Technology – Mechanical Engineering

Jun '10 - Apr '14

Jawaharlal Nehru Technological University-Hyderabad

Hyderabad,Telangana

CERTIFICATIONS/TRAINING

Microsoft Certified: **Azure Fundamentals (AZ-900)**

Microsoft Certified: **Azure Data Fundamentals (DP-900)**

Infosys Certified Software Test Engineer

Data Science Programming Bootcamp – upGrad
